

Economics of Companies—Week 1

The Firm

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I. Defining Companies

Several different types of firms from a legal standpoint;
common forms in the U.S.:

- Corporation—what we usually think of; separate legal entities with rights & responsibilities
- Sole proprietorship—the owner is the firm (e.g., Chris does consulting from the spare bedroom)
- Partnership—e.g., many law firms, consulting firms, and physicians' associations

I. Defining Companies

Corporations

- Limited liability; can sell stock (separation of ownership and control)
- 6.7M returns, \$40T revenues in 2021

Sole proprietorship

- Unlimited liability; income taxed like labor; light regulation
- 29.3M returns, \$1.9 trillion revenues in 2021 [nonfarm]

Partnership

- Some have limited liability (LLC and LLP)
- 4.5M returns, \$12.3 trillion revenues in 2021

I. Defining Companies

We'll consider all these types “firms”

We define a firm as any economic agent that seeks to maximize profits when making a product

We'll assume away ownership vs. control issues (mostly)

II. Companies in Equilibrium

Our descriptions of firms' equilibrium outcomes (size, scope, prices, employment, etc.) generally depend on fundamentals and nature of assumed interactions with other firms

Fundamentals (“primitives”)

- Demand—consumers' tastes
- Technology—production function
- Factor prices—input supply

Interactions

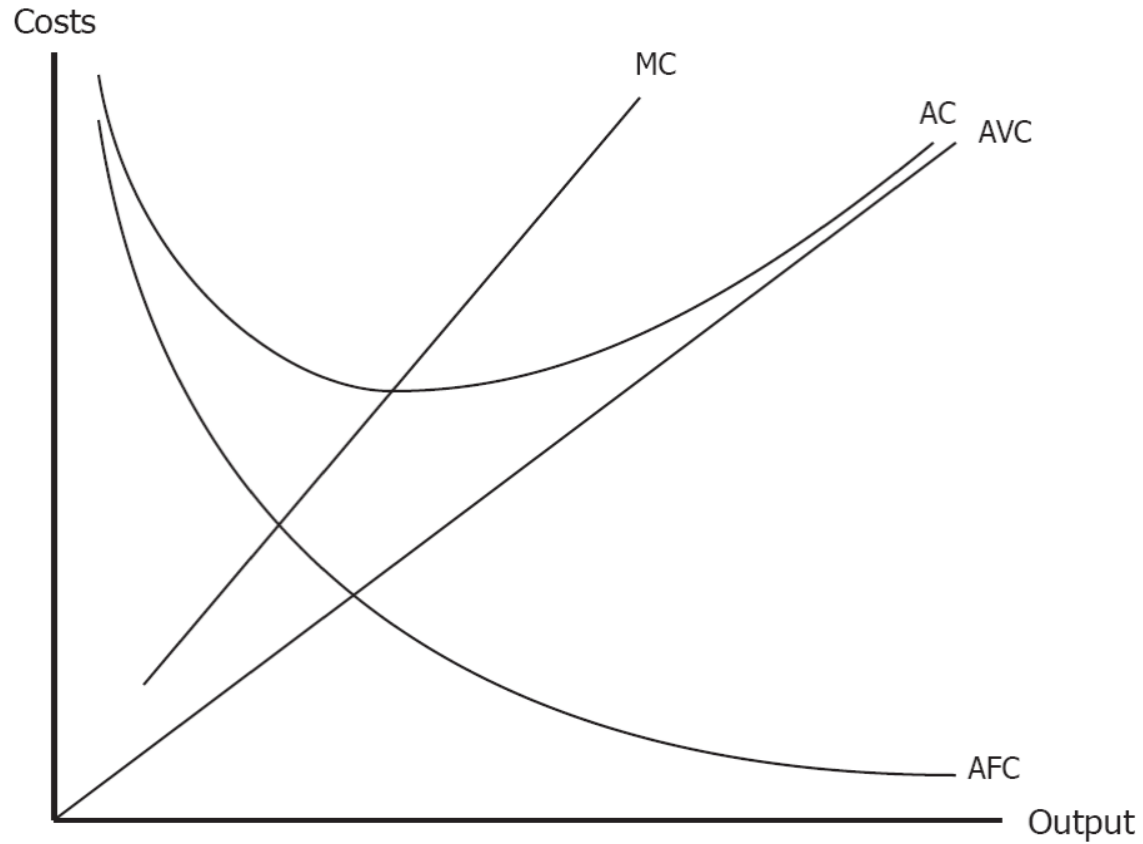
- E.g., Monopoly, Bertrand oligopoly, etc.

III. Costs

The cost curve relates total cost to quantity produced, $C(Q)$

- Important concepts related to the cost curve
 - Fixed costs: must be paid regardless of quantity
 - $FC = C(0)$
 - Average cost: the cost per unit produced
 - $AC = C(Q)/Q$
 - Marginal cost: the cost of producing an additional unit
 - $MC = dC(Q)/dQ$

III. Costs



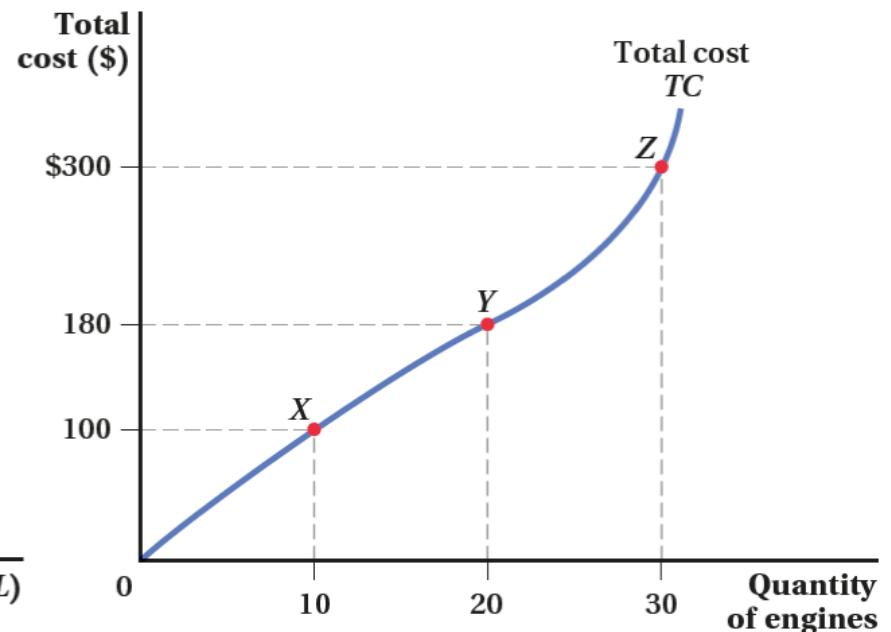
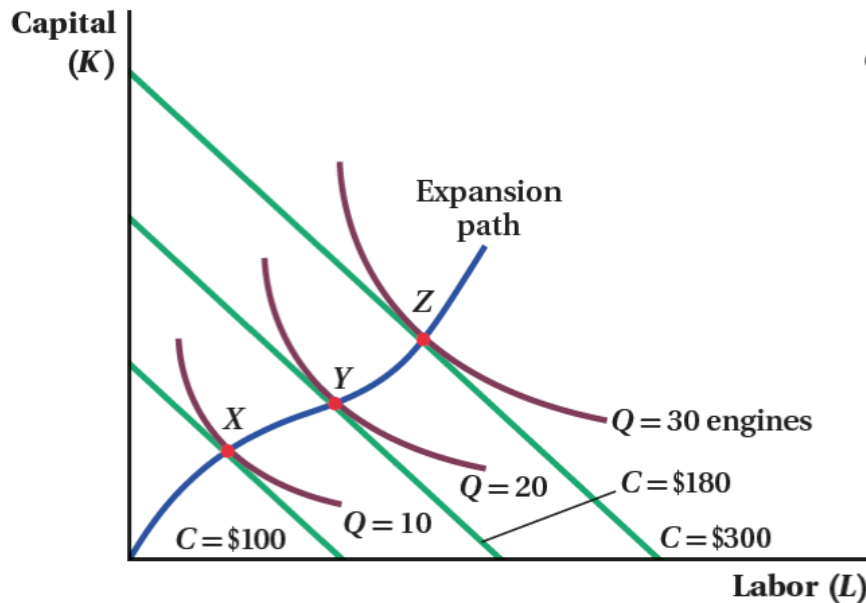
III. Costs

Some things to note:

- MC and AVC need not be linear
- MC must intersect AC at minimum
- AFC must fall in output, as FC spread over more output
- If AC is declining (rising), there are economies (diseconomies) of scale at that quantity
 - This is equivalent to saying economies of scale describe the elasticity of total costs with respect to quantity

III. Costs

Where do costs come from? Remember expansion path, which connects firm's cost-minimizing input choices across output levels (isoquants):



III. Costs

Remember our three fundamentals: demand, technology, factor prices

Isocosts come from factor prices, isoquants from technology

- Isocosts: parallel if factor prices are constant (firm operates in perfectly competitive input markets)
- Isoquants: topographic map of the production function

So a firm's cost function is the outcome of an optimization problem done given its production function

IV. Profit

Profit, π , equals revenue minus costs

- Revenues depend on quantity; costs do too
- So $\pi(Q) = R(Q) - C(Q)$
- If we think firms operate to maximize profits, we have a theory of how they choose Q
- Choose Q to maximize $\pi(Q) = R(Q) - C(Q)$

IV. Profit

- Choose Q to maximize $\pi(Q) = R(Q) - C(Q)$

- First-order condition:

$$\frac{d\pi(Q)}{dQ} = \frac{dR(Q)}{dQ} - \frac{dC(Q)}{dQ} = 0 \quad \Rightarrow \quad \frac{dR(Q)}{dQ} = \frac{dC(Q)}{dQ}$$

- I.e., $MR(Q) = MC(Q)$
- Intuition: if $MR > MC$, profits go up if firm produces more; if $MR < MC$, profits go up if firm produces less

IV. Profit

How does MC change with Q ?

Usual assumptions: MC either flat or rising

- Intuition for flat: replicate a production process
 - Constant returns to scale, or unit economies of scale
- Intuition for rising: capacity constraints of some sort matter, whether physical space or harder to find necessary inputs
 - Decreasing returns to scale, or scale diseconomies

V. Profit under Perfect Competition

How does MR change with Q?

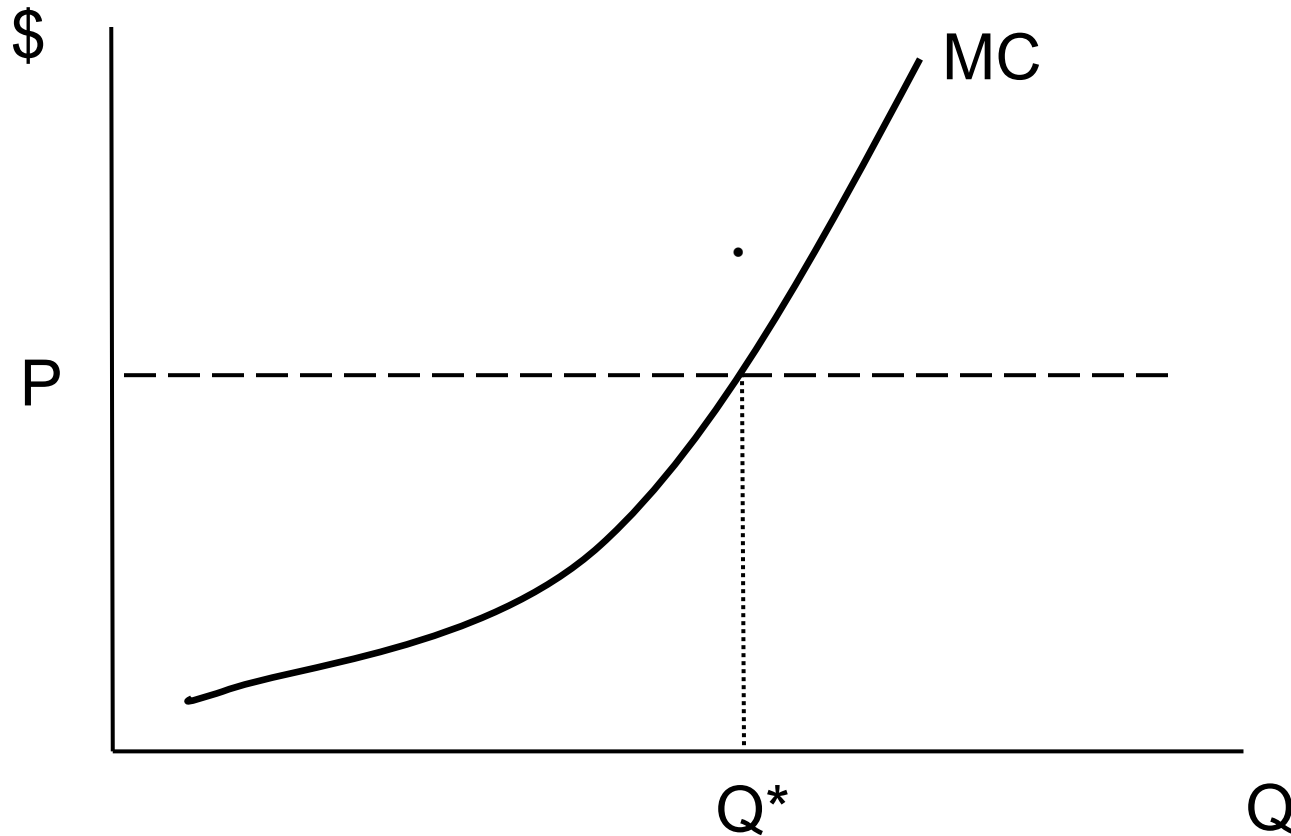
- There is *a lot* of economics dealing with that question
- Very simple, but extreme, case: perfect competition, defined as firms being “price takers”
 - They sell whatever Q at the going market price and do not affect market price with this choice (“atomistic”)
- Under perfect competition, $P(Q) = P$, so $R(Q) = P \cdot Q$ and therefore $dR(Q)/dQ = P$ for all Q
 - I.e., MR is constant and equal to P

V. Profit under Perfect Competition

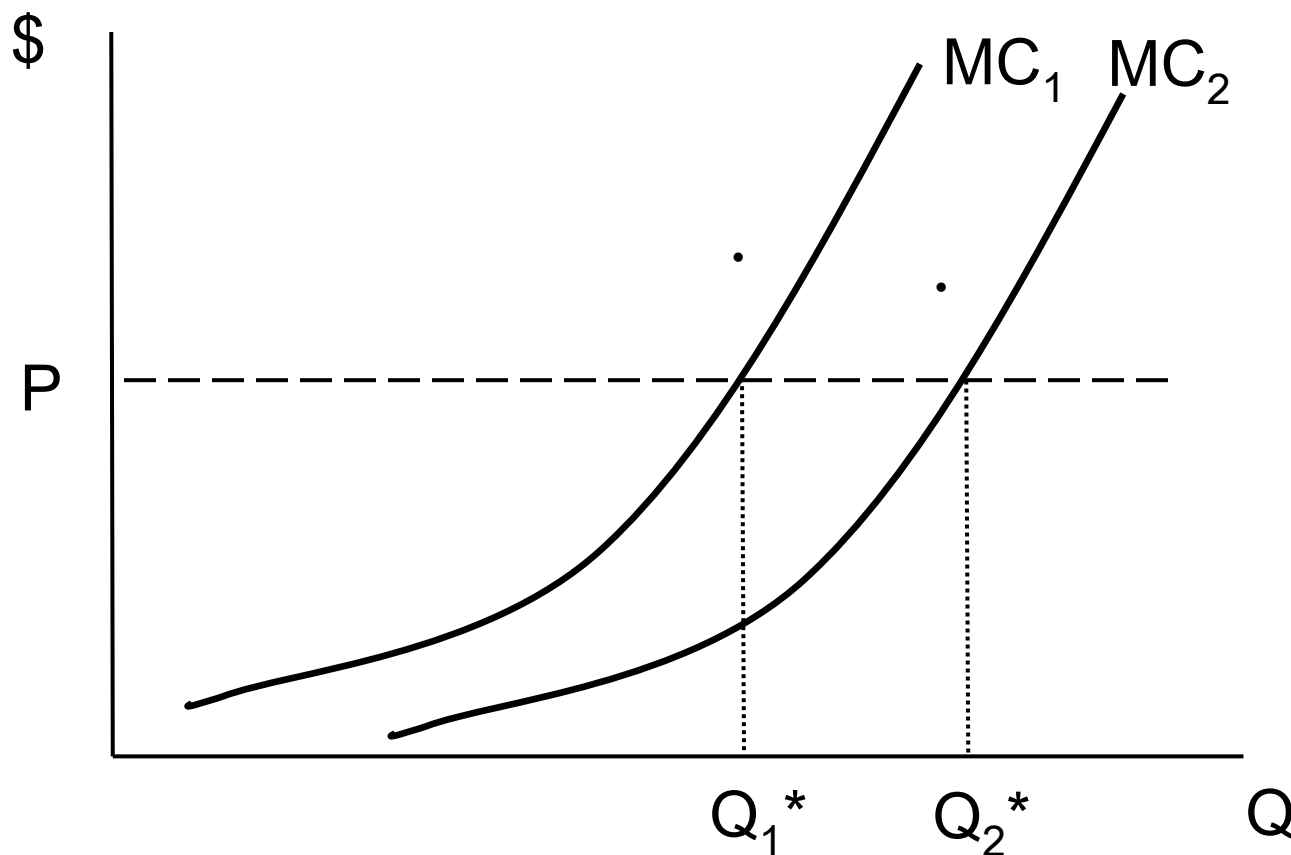
Remember, profits maximized when $MR(Q) = MC(Q)$

- If MR is constant in Q at market price P, MC increases in Q, and $MC(0) < P$, there is some quantity where $MR = MC$
- This is the profit-maximizing quantity
- Because $MR = P$ under perfect competition, profits are maximized here when $P = MC$

V. Profit under Perfect Competition: Graphics



V. Case with Two Firms with Different Costs: Graphics



V. Case with Two Firms with Different Costs

Testable prediction:

- Firms with lower costs (higher productivity and/or facing lower factor prices) should be larger
 - If products are vertically differentiated, this can alternatively be interpreted as implying that firms with higher quality should be larger
 - Here, the “product” is a unit of quality; more efficient firms can produce higher quality at a given price

Case Study: Productivity and Size in Hospitals

Chandra et al. (2016): Are more productive hospitals larger, and do they grow faster?

Compute hospitals' average “risk-adjusted” outcomes from Medicare patient data; compare size and growth across hospitals within market-years

Outcomes: survival, readmission, process-of-care score, patient survey score

Ailments: AMI, congestive heart failure, pneumonia, hip/knee

Case Study: Productivity and Size in Hospitals

Are better hospitals larger (“static allocation”)?

$$\ln(N_h) = \beta_0^s + \beta_1^s q_h + \gamma_M^s + \varepsilon_h^s$$

If so, $\beta_1^s > 0$

Do better hospitals grow faster (“dynamic allocation”)?

$$\Delta_h = \beta_0^d + \beta_1^d q_h + \gamma_M^d + \varepsilon_h^d$$

If so, $\beta_1^d > 0$

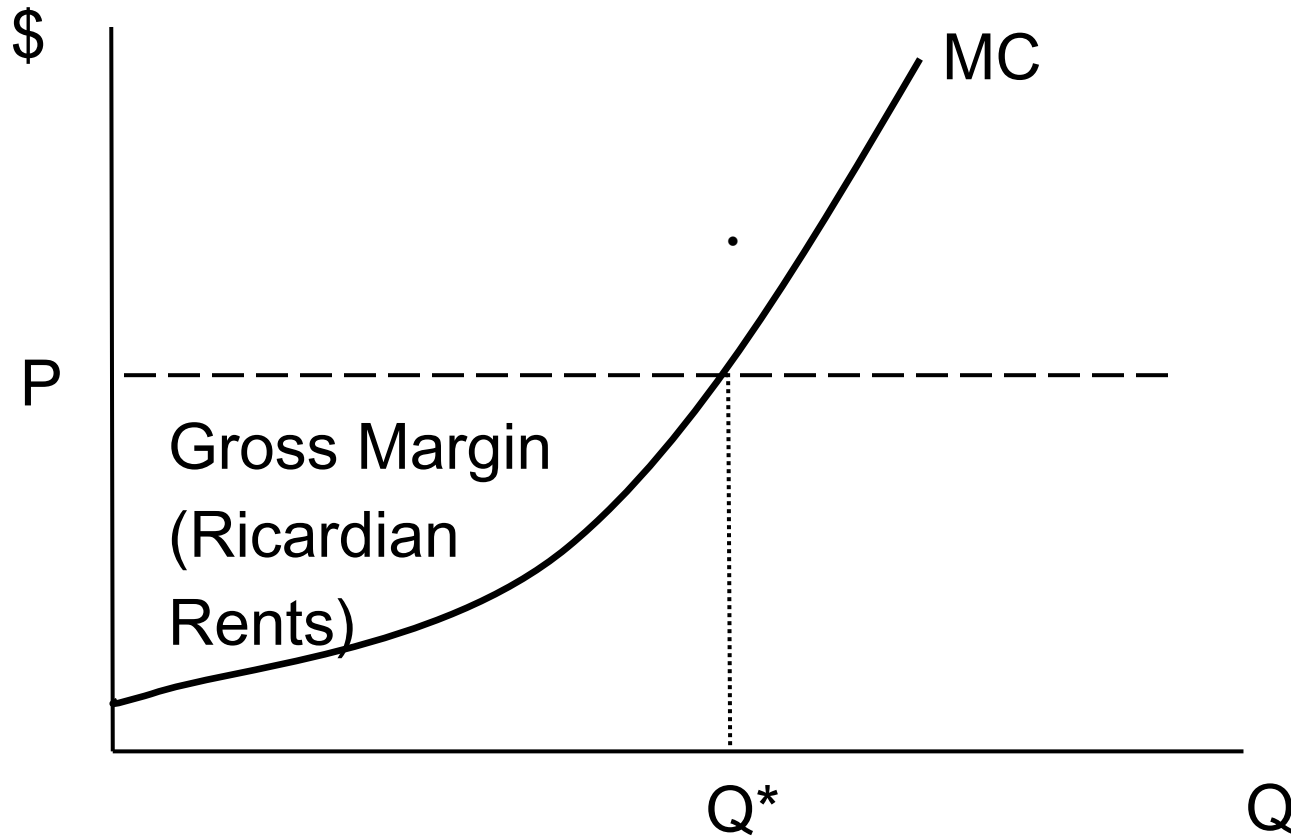
Static Allocation Results

	AMI	CHF	Pneumonia	Hip/Knee
Survival	17.50 (1.00)	15.36 (1.32)	5.14 (0.78)	
Readmission	-9.16 (1.62)	-10.35 (1.78)	0.50 (1.58)	-21.04 (2.03)
Process of care score	0.319 (0.026)	0.332 (0.016)	0.211 (0.015)	
Patient survey score	-0.321 (0.052)	-0.252 (0.038)	-0.210 (0.030)	-0.057 (0.051)

Dynamic Allocation Results

	AMI	CHF	Pneumonia	Hip/Knee
Survival	1.533 (0.379)	0.774 (0.501)	1.220 (0.354)	
Readmission	-1.428 (0.611)	-2.300 (0.651)	-1.138 (0.679)	-1.112 (0.836)
Process of care score	0.048 (0.010)	0.043 (0.009)	0.026 (0.009)	
Patient survey score	-0.065 (0.015)	-0.003 (0.011)	0.007 (0.011)	0.037 (0.022)

V. Profit under Perfect Competition



VI. Profit with Market Power

What if the product market isn't perfectly competitive?

- Now companies **can** affect the market price
- Demand curves slope down, so if they sell more, they must do so at a lower price
- Demand curve: $Q(P)$, where $dQ(P)/dP < 0$
- Inverse demand curve: $P(Q)$, where $dP(Q)/dQ < 0$

VI. Profit with Market Power

$$\text{Revenue} = R(Q) = P(Q) \cdot Q$$

$$MR(Q) = \frac{dP(Q)}{dQ}Q + P(Q)$$

Because $dP(Q)/dQ < 0$, $MR < P$

- Intuition: when a firm increases quantity, it drives down price it sells next unit at, so $MR < P$
- So MR curve is below inverse demand curve

VI. Profit with Market Power

What about slope of MR curve?

$$\frac{dMR(Q)}{dQ} = \frac{d^2P(Q)}{dQ^2}Q + 2\frac{dP(Q)}{dQ}Q$$

- Last term is negative
- Therefore as long as inverse demand curve isn't too convex, $dMR(Q)/dQ < 0$, just like inverse demand

VI. Profit with Market Power

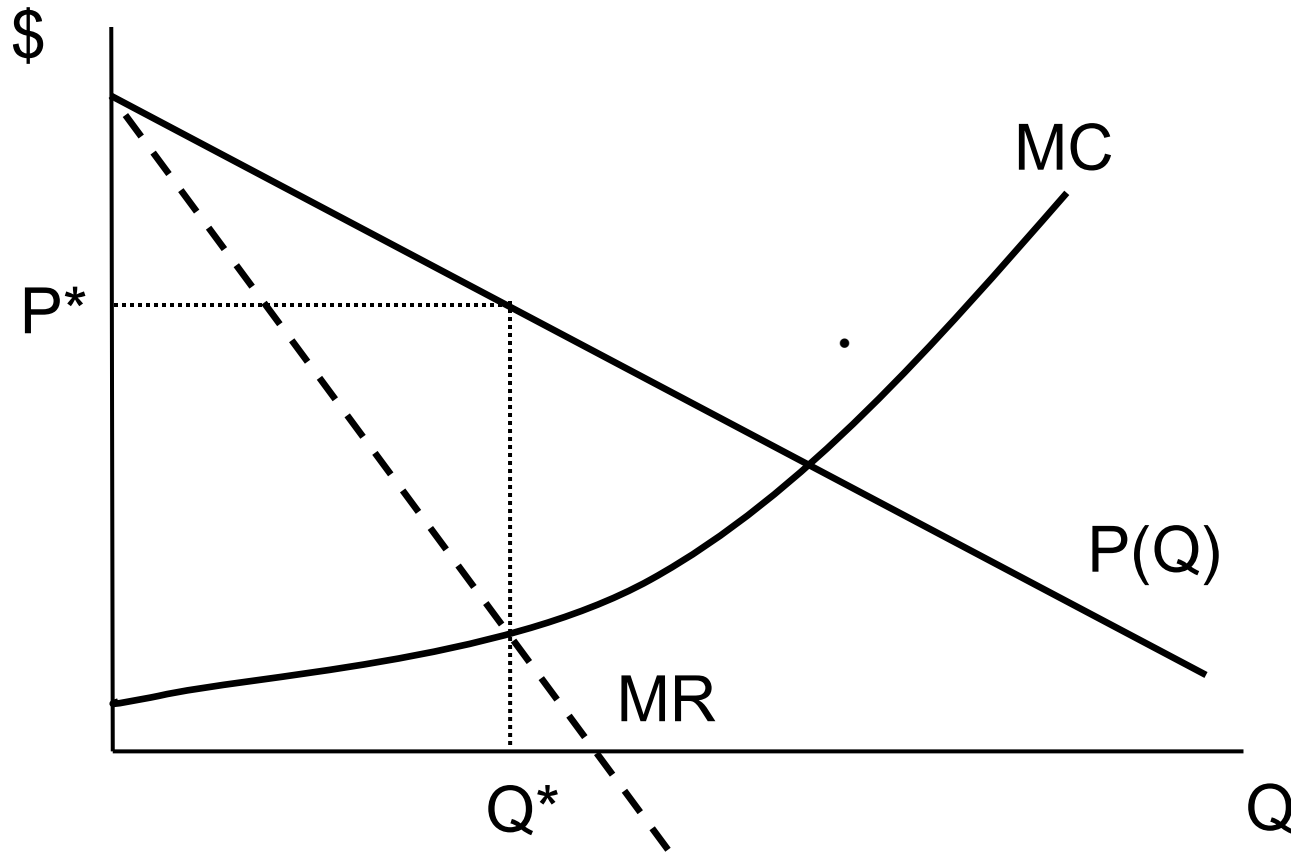
Profit maximization condition is still $MR(Q) = MC(Q)$, but now $MR(Q)$ slopes down and is below inverse demand curve

Again, if MC rises in Q and $MC(0) < P$, there is some quantity where $MR = MC$; that is profit-maximizing quantity

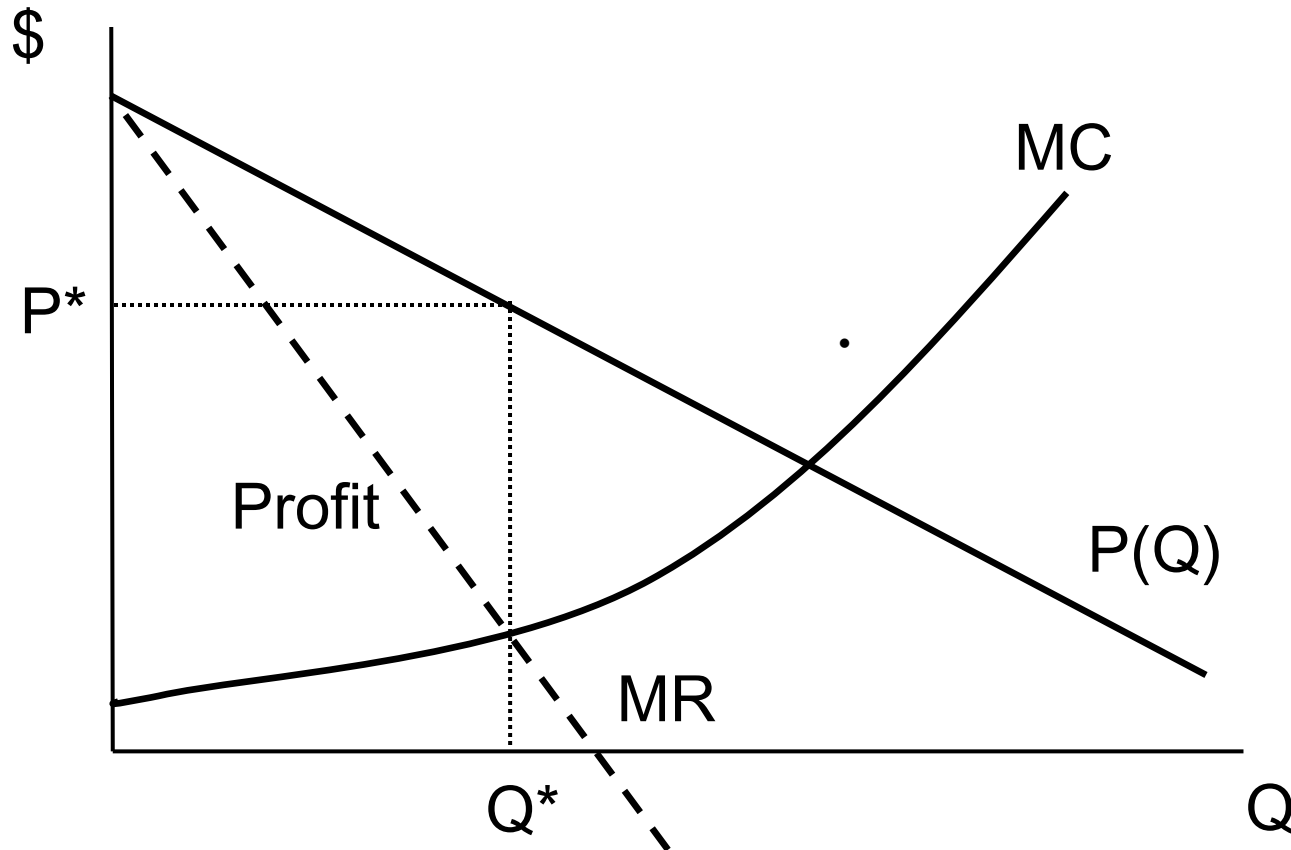
Now $MC(Q)$ can be constant and there's still a profit-maximizing quantity

Profit-maximizing price under market power is just read off demand curve at the profit-maximizing quantity

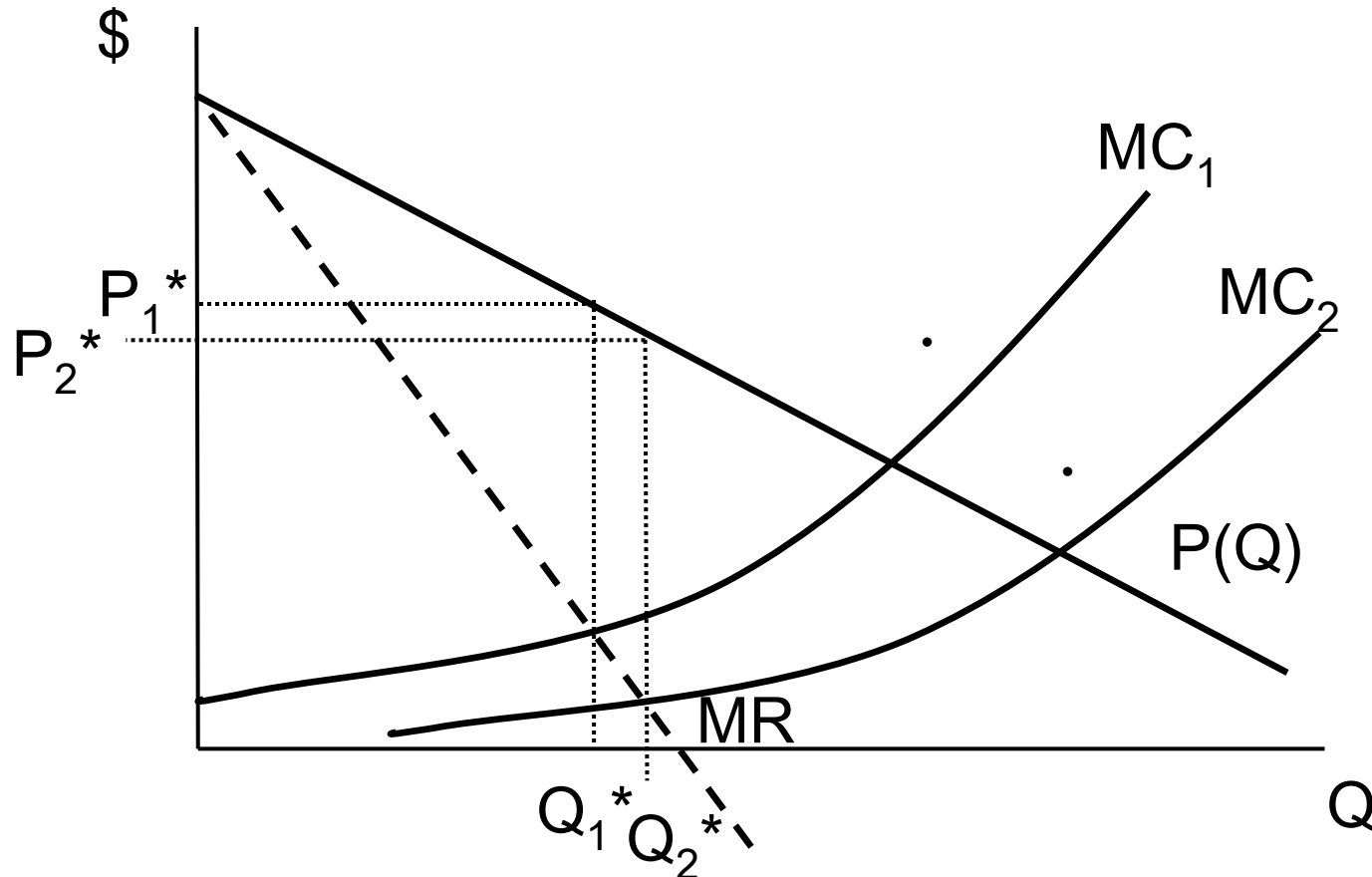
VI. Profit with Market Power: Graphics



VI. Profit with Market Power: Graphics



VI. Profit with Market Power, Two Firms with Different Costs: Graphics



VI. Profit with Market Power: Pricing

Equivalent results viewed through choice of P rather than Q :

$$MR(Q) = \frac{dP(Q)}{dQ}Q + P(Q) = MC(Q)$$

$$P(Q) = MC(Q) + \left(-\frac{dP(Q)}{dQ} \right) Q$$

I.e., profit-maximizing P equals MC plus margin that depends on demand sensitivity

VI. Profit with Market Power: Pricing

Or:

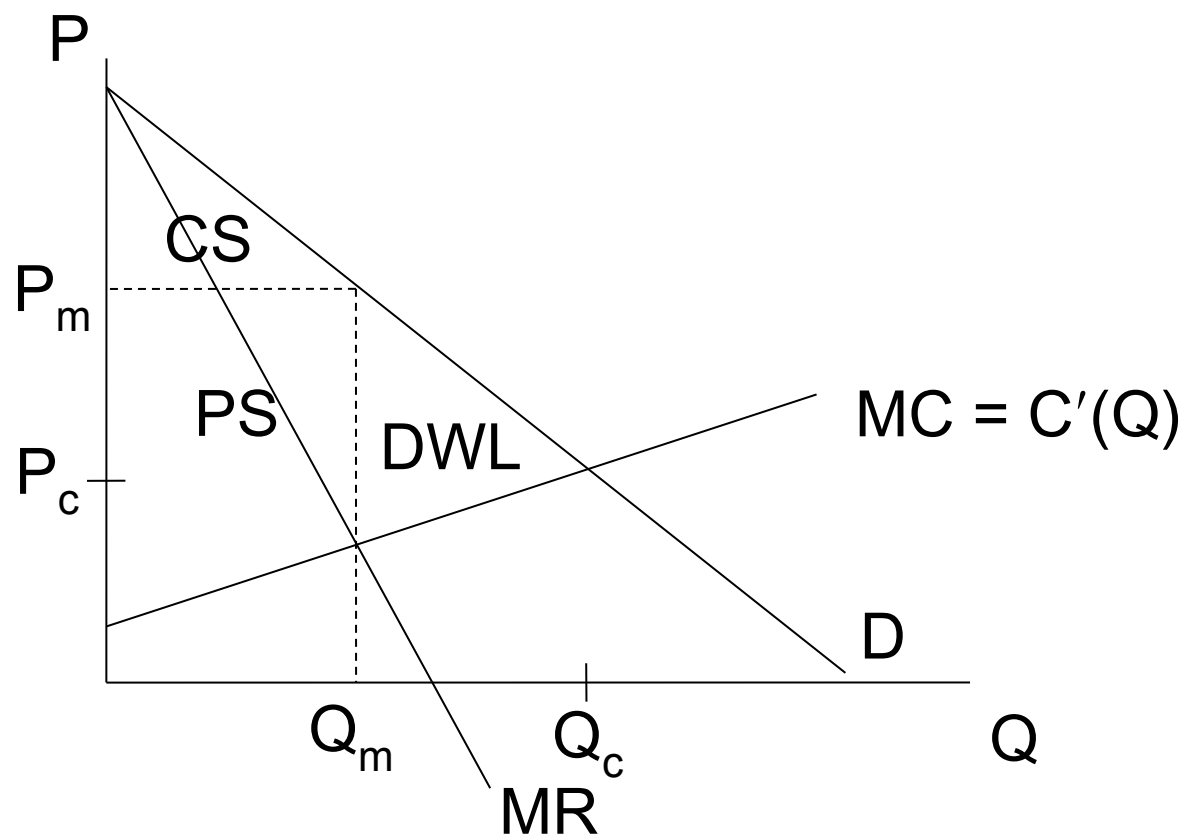
$$P(Q) = MC(Q) - \frac{dP(Q)}{dQ} \frac{Q}{P(Q)} P(Q)$$

$$P(Q) = MC(Q) + \frac{1}{-\epsilon_d} P(Q)$$

$$P(Q) = \left(\frac{|\epsilon_d|}{|\epsilon_d| - 1} \right) MC(Q)$$

I.e., profit-maximizing P equals MC times a markup that depends on demand sensitivity

VII. General Features of Market Power



VII. General Features of Market Power

- Producer surplus (PS) = $P - MC$ summed across units sold; equals profits before fixed costs
- Consumer surplus (CS) = $WTP - P$ summed across units sold
- Deadweight loss (DWL) = lost PS and CS under market power relative to competitive outcome
 - Captures units with $WTP > MC$ that aren't made but should be from social perspective
 - Part of PS in monopoly can be a social loss too if it is spent on wasteful rent-preserving activities

VII. General Features of Market Power

- More market power does not necessarily mean more DWL
- When demand is steeper, $WTP - MC$ is higher for unsold units, but there are fewer of them
- In fact, total market power (perfectly inelastic demand) has no DWL
- Though all social surplus is producer surplus (distribution/equity concern)

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